PROJECT DESCRIPTION

- Data:

Data source: WVS7 Questionnaire USA 2017 English version
The dataset contains 394 variables and 2,596 cases.

- The target goal (question to answer):

Does the dataset support the following social psychology theories (Social Identity Theory, Existential Security Theory, Relative Deprivation Theory, Just-World Belief Theory, Trust Spillover Hypothesis)?

- The people you would recruit to answer that question:

People who are experienced in the field of statistics or data visualization.

- Tasks:

The kinds of measures you would use to answer your data (e.g., insight depth, use cases, accuracy) and what these measures would tell you about the core question?

a) Insight depth
   Number of patterns discovered, number of unique discoveries per dataset. Helps to discover explanatory power of findings.
b) Actionable use cases
   What actionable recommendations can be generated from these data vizes?
c) Theoretical Grounding:
   Is the theory supported by the data vizes?

- How would you instantiate those methods (i.e., what would your participants do?)

Explore the data visualizations to generate new knowledge.

- What criteria would you use to indicate that your visualization was successful

a) Pre-attentive Processing: Can a viewer identify the main takeaway in under 5 seconds?
b) Theoretical Grounding: Does the visualization clearly show the relationship defined by the theory?
c) Insight Depth: Does the visualization reveal a pattern that wasn't obvious from the raw data?
d) Actionable Use Cases: Does the visualization allow to form a new hypothesis?

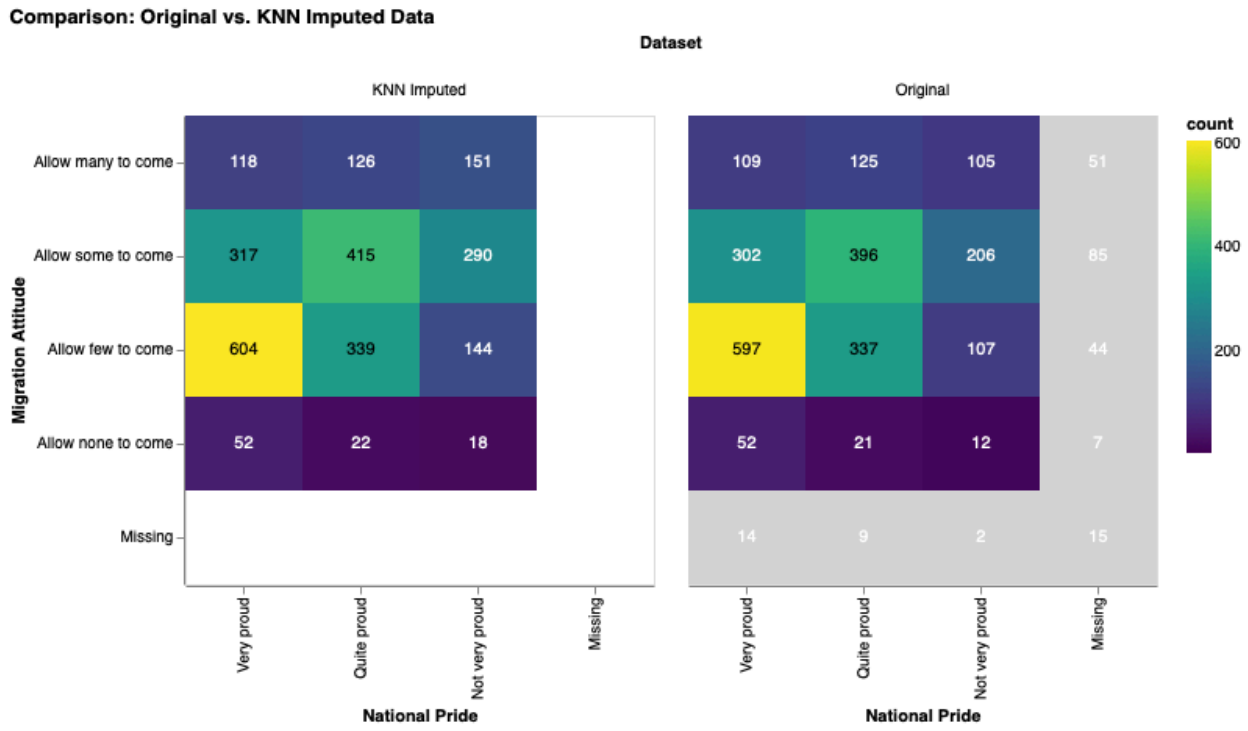
DATA ANALYSIS (Link to Colab):
https://colab.research.google.com/drive/1iBePDm5949UFAccdRMsK6rQIBHF0jMQC?usp=sharing
DATA VISUALIZATIONS:

1. Social Identity Theory
   People categorize themselves into "ingroups" (us) and "outgroups" (them), creating positive distinctiveness for their own group, which can lead to prejudice, discrimination, and intergroup conflict. © EBSCO

WVS variables:

- National pride
- Willingness to have immigrants as neighbors / confidence in foreigners
- Confidence in institutions / support for democracy

Visualization:
Bivariate Heatmap, visualizing the correlation between ingroup favoritism (National Pride) and outgroup prejudice (Migration Attitude) to confirm social identity theory.
Justification:
Shows statistical significance.
The grey variable for missing data.
Side-by-side comparison of raw data against KNN-imputed data.

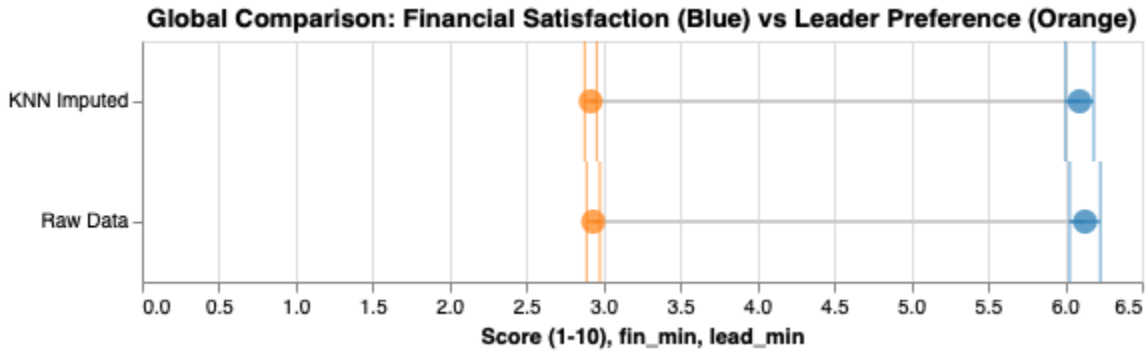


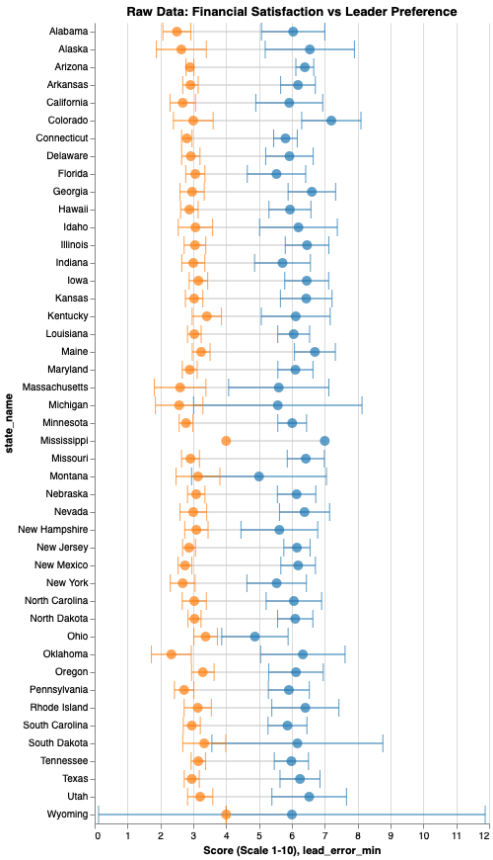
2. Existential Security Theory
   high levels of security (economic, physical, and safety) lead to secularization and democratic values, whereas high levels of existential insecurity—driven by poverty, disease, or economic instability—lead to a greater demand for religiosity and authoritarian leadership. © sociologytwynham.com

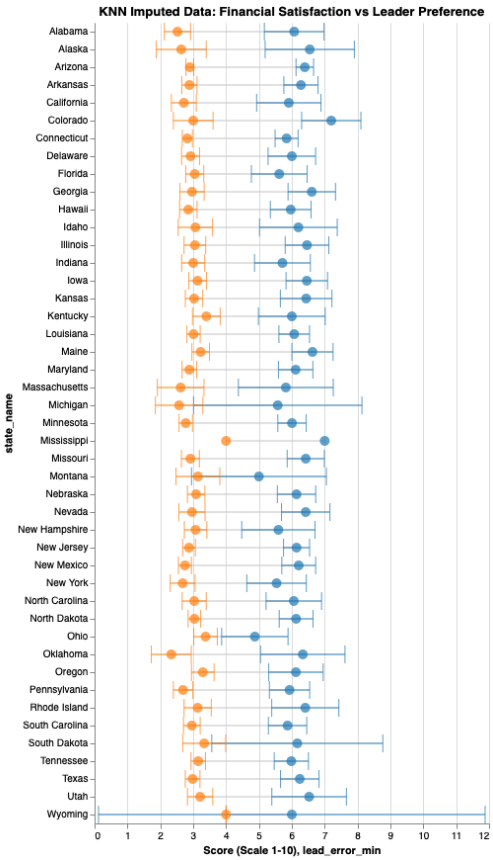
WVS variables:

- Financial satisfaction / perceived economic insecurity
- Approval of strong leader / military rule

Visualization:
Global comparison & state-level variance using dot plots with error bars.
Justification:
Clarity of correlation by placing both variables on the same x-axis scale.
The inclusion of error bars.
Side-by-side comparison of raw data against KNN-imputed data.







3. Relative Deprivation Theory

   Discontent and social movements arise not from absolute poverty, but from the subjective perception that one is worse off compared to others or one's own expectations. © LSE

WVS variables:

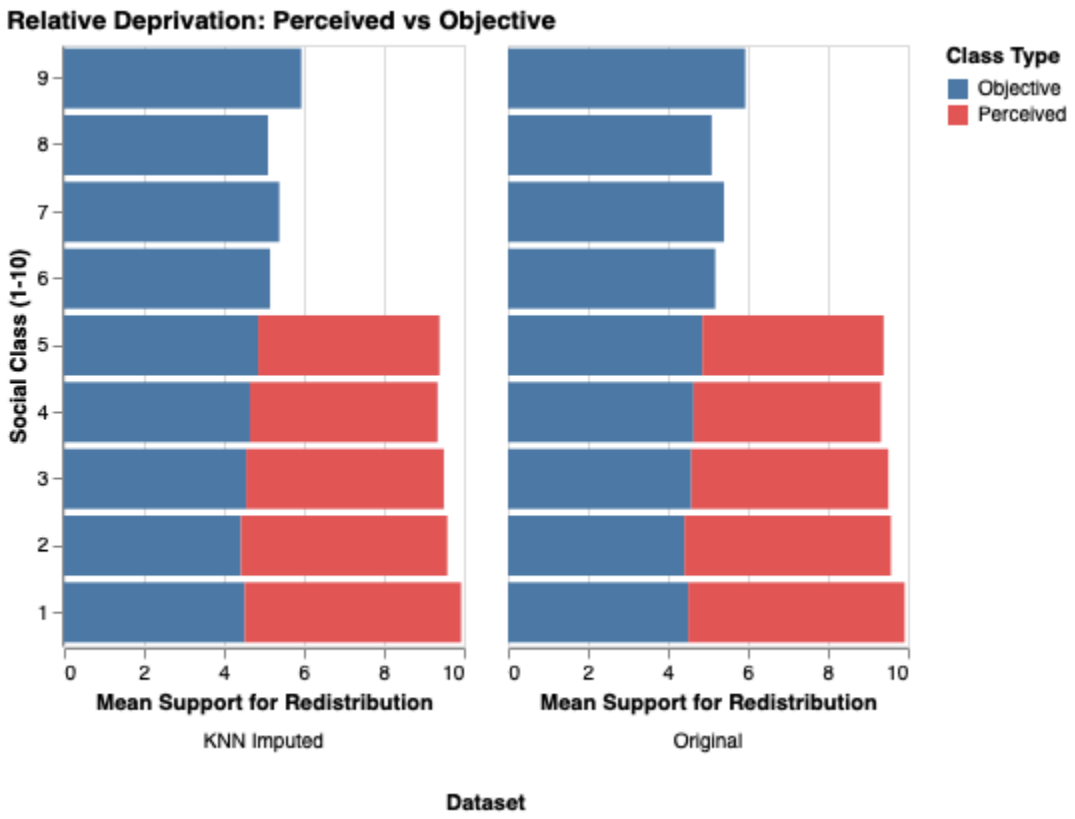
- Subjective social class / perceived unfairness
- Support for redistribution / confidence in institutions

Visualization:

Faceted Stacked Bar Chart. Self-perception as a lower-class carries an additive effect on political demands.

Justification:

The KNN imputed chart is identical to the original chart, suggesting that the missing data was likely "missing at random"

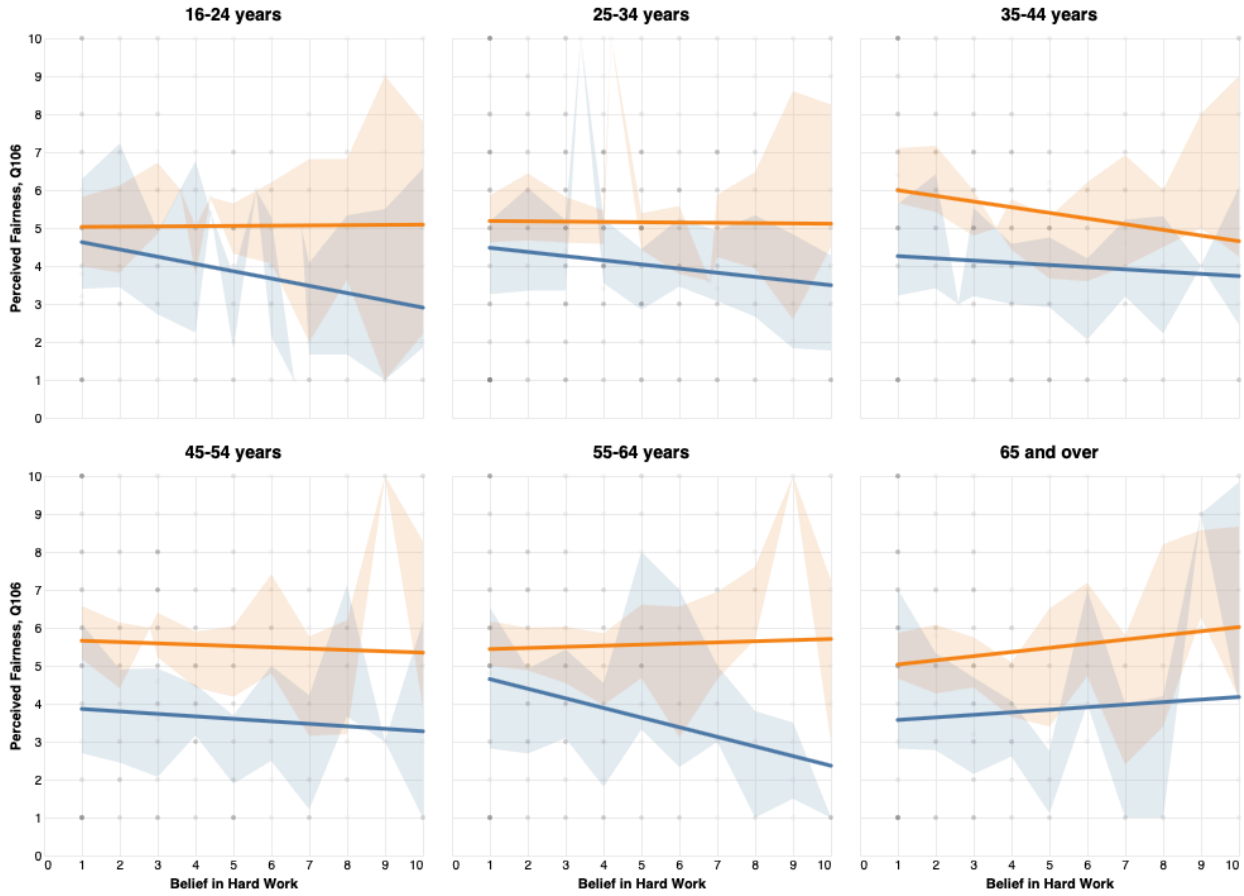


4. Just-World Belief Theory
   The cognitive bias or belief that the world is fair and people get what they deserve, which helps individuals make sense of events around them.

WVS variables:

- “Hard work usually brings success”
- Views on income inequality, support for welfare, confidence in institutions

Visualization:
Faceted scatter plot with linear regression lines and confidence intervals
Justification:
The data is split into a grid of subplots based on a categorical variable (age groups), allowing for comparison across cohorts.
The underlying gray dots represent individual data points.
The blue and orange lines represent the correlation between belief in hard work and perceived fairness.
The shaded areas around the regression lines represent the variance in the data.

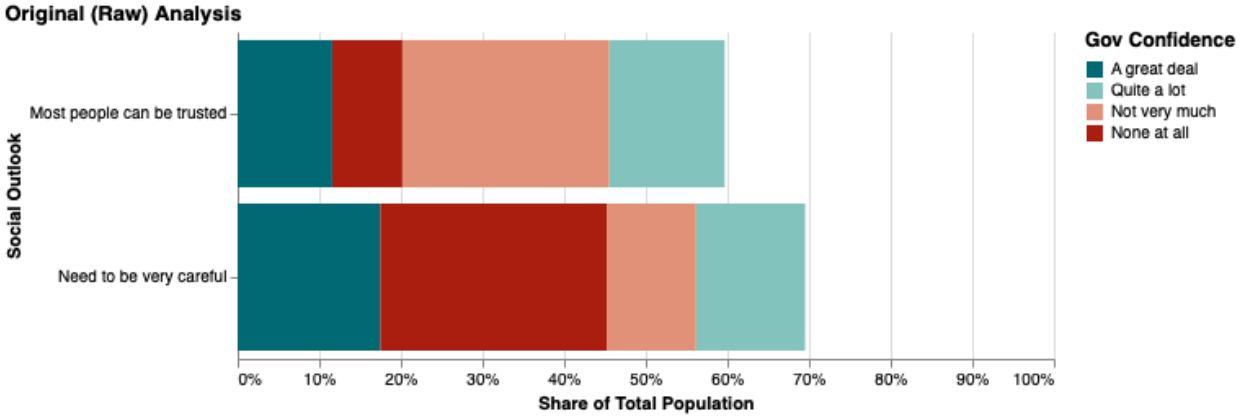


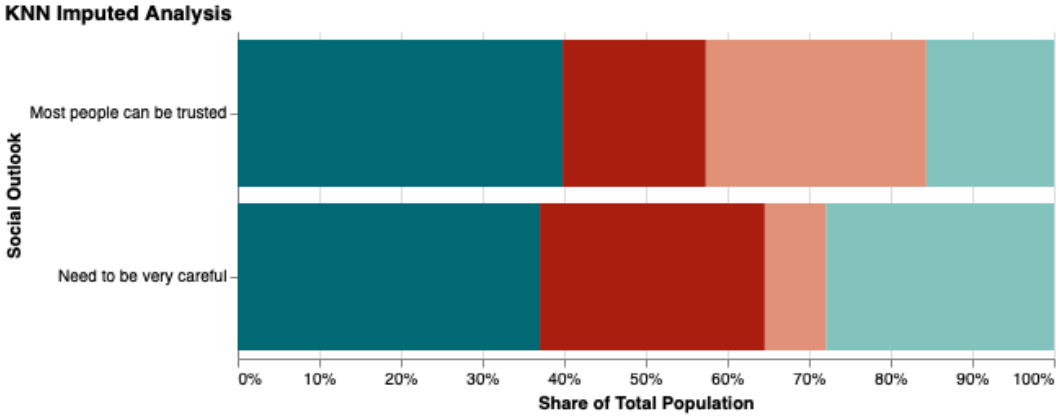
5. Trust Spillover Hypothesis

WVS variables:

- Generalized trust (“Most people can be trusted”) Q57
- Confidence in government

Visualization:]
Stacked Bar Chart
Justification:
There is a significant amount of missing data. Non-response bias could be skewing the results.
By using KNN to fill gaps, the bars now reflect 100% of the population.





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CONCLUSION

- Discussion of your final evaluation approach, including the procedure, people recruited, and results & A synthesis of findings, including what elements of your approach worked well and what elements you would refine in future iterations.

People recruited:
Friends & Family

Final evaluation approach:

1. Social Identity Theory
Pre-attentive Processing: The bright yellow cell draws attention.
Theoretical Grounding:. Maps "ingroup favoritism" (National Pride) against "outgroup prejudice" (Migration Attitude).
Insight Depth: Main Insight: People who are "Quite proud" have a significant lean toward restricting migration.
Actionable Use Cases: Supports a hypothesis that high national pride is a primary predictor for restrictive migration views in the USA.

2. Existential Security Theory
Pre-attentive Processing: The state-level plots take longer to fully understand the trend due to the volume of labels .
Theoretical Grounding: Tests the link between economic security and authoritarianism.
Insight Depth: The error bars show variance that raw means would hide.
Actionable Use Cases: Allows to identify (outliers) that deviate from the theory for further study.

3. Relative Deprivation Theory
Pre-attentive Processing: The red blocks clearly show an additive effect on political demands across social classes.
Theoretical Grounding: It visualizes the subjective perception vs. objective social class.
Insight Depth: Highlights that the KNN-imputed data is identical to the original, validating that missing data was likely random.
Actionable Use Cases: subjective class identity has a more consistent impact on redistribution support than objective metrics.

4. Just-World Belief Theory
Pre-attentive Processing:
The regression lines indicate the direction of the relationship for each age cohort.
Theoretical Grounding:
It compares the "Belief in Hard Work" against "Perceived Fairness" across different life stages. +
Insight Depth:
The faceted approach reveals that the correlation may weaken in older cohorts (65+).
Actionable Use Cases:
Suggestion of a new hypothesis: Just-world beliefs may be more influenced by life experience and age than a static cultural trait.

5. Trust Spillover Hypothesis
Pre-attentive Processing: Bars allow for an instant comparison of trust levels.
Theoretical Grounding:
Linking generalized trust to government confidence.
Insight Depth:
Using KNN to fill gaps reveals how non-response data might have skewed the result

Synthesis of Findings

What Worked:

1) Comparative Analysis: The side-by-side comparison of Original vs. KNN Imputed data. It provides transparency on how data cleaning affects the final result.
2) Theoretical Alignment: Each chart was chosen to represent a psychological theory (e.g., Heatmaps for Identity, Dot Plots for Security).

Possible Refinements:

1) Reducing Cognitive Load: The state-level dot plots are quite "busy". In the future, the solution may be to use an interactive hover element.
2) Color Theory: While color was used to distinguish variables, using a unified color palette across all five theories could help create a more cohesive report.
3) More descriptive labels are required

APPENDIX:

SURVEY:

1. Pre-attentive Processing

- **Question:** "Looking at this chart for 5 seconds, what is the single most important trend or data point you notice?"

2. Theoretical Grounding

Question: "On a scale of 1–10, how clearly does this visualization illustrate the relationship defined by [Theory Name]?"

3. Insight Depth

- **Question:** "What pattern do you see that you would have missed looking at a standard data table?"

4. Actionable Use Cases

- **Question:** "What new hypothesis would you propose for further study based on this data visualisation?"

5. Improvements:

- "Were the side-by-side comparison of KNN-imputed data and raw data helpful for interpretation?"
- "Did the color choices or labeling help or hinder your interpretation?"
- What suggestions could you provide?

REPLIES:

REPLY #1:

I

1 Yellow square is distinct.

2 8

3 The viz has yellow and green squares that may be missed in raw data

4 I'd research data about what migration attitude immigrants have

5 Color worked well!

II

1 Orange dots are always on the left, blue on the right

2 7

3 Authoritarism is not supported by financially satisfied people

4 I'd compare US to less democratic countries

5 The use of contrast colors is a good choice

III

1 No red lines above 6

2 4

3 Perceived social class distribution

4 Conduct a new survey and find more people who perceive themselves as wealthy

5 Stacked bars look confusing

IV
1 Yellow areas
2 9
3 Only people over 65 belief in fairness
4 Checking data by state
5 Add labels to the graph itself

V
1 People don’t have enough confidence in government
2 6
3 The missing data represenation
4 Check trust in other agencies, not only the government
5 Green add to green, red and yellow together

REPLY #2
1 Yellow square.
2 8
3 I wouldnt find the link between "ingroup favoritism" and "outgroup prejudice"
4 N/A
5 I liked the contrast between colors

II
1 Everything is evenly distributed but too overcrowded
2 6
3 Patterns for all states look the same
4 N/A
5 Add the legend to the second graph (from the first one)

III
1 Red lines
2 8
3 Respondents perceive themselves as low/ middle class
4 That self-perception plays a greater role than objective status
5 Change the colors to more accessible

IV
1 Correlation lines
2 8
3 Error areas & that blue line is lower than yellow
4 Check if immigrants have the same distributions as non-immigrants
5 Everything is ok to me, the background maybe unnecessary

V
1 People who trust the government tend to trust people more
2 8
3 Missing values
4 Check gender, and age distribution
5 The values should be grouped logically by color and not divided (in my opinion)

REPLY #3
1 First yellow, then purple color.Then the absence of gray color on the first image.
2 9
3 I would notice that KNN values worked pretty well
4 Adding the bars showing totals?
5 The missing values are handled well

II
1. Financially satisfied people don’t need a strong leader. Also, the use of error bars draws attention
2 8
3 I might have missed the error bars if it wouldn’t visualized
4 I’d combine states into region (maybe there are not enough responses in each state, that’s why the error bars are large)
5 The error bars for Wyoming are concerning, but i liked that there is a comparison for the whole population as well

III
1 Red bars
2 7
3 The lack of number 10
4 Support for redistribution lessens with the income
5 Maybe another chart type, but I’m not sure which one

IV
1 Two correlation lines & age distribution
2 10
3 Age group of 65yo vs other age groups
4. Check, if it is influenced more by age or life experience?
5 The use of error bars and the background was on point!

V
1 A lot of missing values
2 8
3 I think the KNN computed values are not accurate enough (but I may be wrong)
4 Do demographic factors play a role?
5 The representation of missing values is helpful

<u>AI Usage Archive:</u>

- Brainstorming what theories could be analyzed based on wvs dataset.**OpenAI. (2026). ChatGPT - Chat #1**
- Asking for help with generating a questionnaire for data viz evaluation. **Google (2026). Gemini - Chat #2**
- Asking for help with programming syntax and general programming questions that are not directly related to the solution of the assignment in question. **Google (2026). Gemini**
- Check spelling & grammar. **Grammarly (2026). Grammarly.**

BRAINSTORMING STAGE:
CHAT #1

USER:
what questions could i ask based on the dataset World Values Survey
what research questions could i ask and what for based on sci frameworks

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AI:
Ooo okay. Now we're leveling up 🧠✨
You're not just asking "what can I analyze?" — you're asking:

> What research questions can I ask from **scientific frameworks**, and what would they help me test or understand?

Since you're working with **World Values Survey (WVS)** data, we can ground your questions in established social science theories.
Below I'll structure this as:

1. 📚 Framework
2. ❓ Research Question
3. 🎯 What it tests
4. 📊 How you'd test it with WVS

1️⃣ **Modernization Theory**
(Ronald Inglehart, post-materialism)
📚 **Core Idea**
As societies become economically secure, values shift from survival → self-expression.

❓ **Research Question**
Do higher-income or more economically secure individuals show stronger self-expression values (e.g., tolerance, autonomy, gender equality)?
🎯 **What it tests**
Whether economic security drives cultural value change.
📊 **How with WVS**

- Independent variable: income satisfaction, financial security
- Dependent variables: tolerance, democracy support, freedom importance
- Model: regression or cohort analysis

Advanced:

- Compare younger vs older cohorts → generational shift

2️⃣ **Social Capital Theory**
(Robert Putnam)
📚 **Core Idea**
Trust and civic engagement create better social and political outcomes.

❓ **Research Question**
Does interpersonal trust predict life satisfaction and support for democracy?

🎯 **What it tests**
Whether trust functions as a form of “social infrastructure.”
📊 **With WVS**
- “Most people can be trusted”
- Life satisfaction
- Confidence in institutions
- Participation in organizations

Possible extension:
Does trust mediate the relationship between income and happiness?
That would be very strong analytically.

3️⃣ **Self-Determination Theory (SDT)**
(Edward Deci & Richard Ryan)
📚 **Core Idea**
Well-being depends on:
- Autonomy
- Competence
- Relatedness

❓ **Research Question**
Does perceived freedom of choice (autonomy proxy) predict life satisfaction more strongly than income?
🎯 **What it tests**
Psychological vs material determinants of happiness.
📊 **With WVS**
- Freedom of choice
- Life satisfaction
- Income
- Social relationships

You could test:
Autonomy > income in predicting happiness?
Very portfolio-ready.

4️⃣ **System Justification Theory**
(John Jost)
📚 **Core Idea**
People justify existing inequalities, even if they disadvantage them.

❓ **Research Question**
Do lower-income individuals still support income inequality if they believe hard work leads to success?
🎯 **What it tests**
Internalization of system legitimacy.
📊 **With WVS**
- Income level
- Belief that hard work leads to success
- Views on income inequality

This becomes a psychological-political study.

5️⃣ **Schwartz Value Theory**
(Shalom Schwartz)
Since you’ve explored Schwartz before 👀
📚 **Core Idea**
Values are structured along dimensions:
- Openness to change vs conservation
- Self-enhancement vs self-transcendence

❓ **Research Question**

Do individuals high in conservation values show lower tolerance toward minority groups?
🎯 **What it tests**
Value-attitude consistency.
📊 **With WVS**
- Tradition importance
- Security importance
- Attitudes toward immigrants / LGBTQ+

Advanced:
Factor analysis to reconstruct value dimensions.

6️⃣ **Terror Management Theory**
(Sheldon Solomon)
📚 **Core Idea**
When existential insecurity rises, people cling to tradition and authority.

❓ **Research Question**
Do individuals who feel unsafe or economically insecure show stronger authoritarian preferences?
🎯 **What it tests**
Security → conservatism hypothesis.
📊 **With WVS**
- Perceived safety
- Economic insecurity
- Preference for strong leaders
- Traditional values

7️⃣ **Institutional Trust & Legitimacy Framework**
❓ **Research Question**
Is trust in institutions more strongly related to democracy support than economic satisfaction?
🎯 **What it tests**
Institutional legitimacy as a democratic stabilizer.

8️⃣ **Cultural Tightness-Looseness Theory**
(Michele Gelfand)
📚 **Core Idea**
Tight cultures enforce norms strongly; loose cultures tolerate deviation.

❓ **Research Question**
Are countries with high emphasis on tradition and authority less tolerant of minority groups?
🎯 **What it tests**
Norm enforcement vs pluralism dynamics.

Now Let's Get Very Strategic For You
Given:
- You like psychology
- You like value systems
- You want analytical depth
- You're thinking about research / data portfolio

Here are 3 high-level cohesive project directions:

🌍 **Project 1: "What Really Drives Happiness?"**
Framework:
- Self-Determination Theory
- Social Capital Theory

Test:
Income vs Autonomy vs Trust → Life Satisfaction

Why:
Strong for policy + analytics + psychology.

🧠 **Project 2: "Security vs Freedom: A Global Value Tradeoff"**
Framework:
- Modernization Theory
- Terror Management Theory

Test:
Does insecurity increase support for authority and tradition?
Why:
Deep sociological angle.

🔥 **Project 3: "Do Values Predict Political Attitudes?"**
Framework:
- Schwartz Value Theory
- System Justification Theory

Test:
Do core values structure support for inequality, democracy, minority rights?
Why:
Perfect if you're writing a social commentary book.

USER:
Please write 15 additional research questions:
AI:
Love this direction. Let's push it deeper and more structurally rigorous 🧠✨
Below are **15 additional research questions**, each grounded in a scientific framework and clearly stating what they help you test.

1️⃣ **Social Identity Theory**
(Henri Tajfel)
RQ1:
Does strong national pride predict lower tolerance toward immigrants?
🎯 Tests whether in-group identification reduces openness to out-groups.

2️⃣ **Relative Deprivation Theory**
(Ted Gurr)
RQ2:
Do individuals who perceive themselves as worse off than others show stronger support for redistributive policies?
🎯 Tests whether perceived inequality (not actual income) drives political preferences.

3️⃣ **Political Efficacy Theory**
RQ3:
Does perceived political efficacy (belief that one's voice matters) predict civic participation more strongly than education level?
🎯 Tests psychological vs structural drivers of participation.

4️⃣ **Cultural Backlash Theory**
(Pippa Norris)
RQ4:
Are older cohorts more likely to support authoritarian leadership in societies experiencing rapid cultural change?
🎯 Tests whether modernization produces generational backlash.

5️⃣ **Post-Materialist Value Shift**
(Ronald Inglehart)

RQ5:
Do younger generations prioritize environmental protection over economic growth more than older generations?
🎯 Tests generational value transformation.

6️⃣ **Institutional Legitimacy Theory**
RQ6:
Does trust in courts and parliament predict compliance with democratic norms more strongly than satisfaction with the economy?
🎯 Tests whether democracy stability is value-based or performance-based.

7️⃣ **Just-World Belief Theory**
(Melvin Lerner)
RQ7:
Do individuals who believe that success depends on hard work show lower support for welfare policies?
🎯 Tests moralization of inequality.

8️⃣ **Secularization Theory**
RQ8:
Is religiosity negatively associated with acceptance of divorce, abortion, or LGBTQ+ rights across cultural contexts?
🎯 Tests whether secularization predicts moral liberalization.

9️⃣ **Existential Security Theory**
RQ9:
Does perceived financial insecurity increase preference for a “strong leader who does not have to bother with parliament”?
🎯 Tests security-authoritarianism link.

🔟 **Trust Spillover Hypothesis**
RQ10:
Does interpersonal trust spill over into institutional trust?
🎯 Tests whether micro-level trust scales up to macro-level political legitimacy.

1️⃣1️⃣ **Human Development Theory**
RQ11:
Does education level moderate the relationship between income and tolerance?
🎯 Tests whether cognitive development reduces economic-based prejudice.

1️⃣2️⃣ **Life Course Theory**
RQ12:
Do major life events (e.g., unemployment, divorce) predict long-term shifts in political ideology?
🎯 Tests whether values are stable or reactive.

1️⃣3️⃣ **Cultural Tightness-Looseness**
(Michele Gelfand)
RQ13:
Do societies emphasizing obedience and tradition exhibit lower acceptance of social deviance?
🎯 Tests norm enforcement at cultural level.

1️⃣4️⃣ **Social Dominance Theory**
(Jim Sidanius)
RQ14:
Do individuals who prioritize hierarchy values show lower support for gender equality?
🎯 Tests dominance orientation in value systems.

1️⃣5️⃣ **Happiness Economics**
RQ15:
Does relative income position (income compared to country average) predict life satisfaction more strongly than absolute income?
🎯 Tests whether happiness is comparative rather than absolute.

BRAINSTORMING STAGE:
CHAT #2

USER:
please provide 5 different questionnaire questions that would result in answer to the following questions & evaluations of data viz graphs: Pre-attentive Processing, Theoretical Grounding, Insight Depth, Actionable Use Cases, Refinement recommendations in future iterations
like survey questions about the data visualizations

AI:
You need the specific survey questions you would ask participants to evaluate the **effectiveness of your visualizations** based on the four criteria you defined.

1. Pre-attentive Processing (The "5-Second Rule")
Goal: Determine if the main takeaway is identified instantly.
- **Question:** "Looking at this chart for 5 seconds, what is the single most important trend or data point you notice?"

2. Theoretical Grounding
Goal: Check if the visual accurately represents the theory.
- **Question:** "On a scale of 1–10, how clearly does this visualization illustrate the relationship defined by [Theory Name]?"

3. Insight Depth
Goal: See if the viewer discovers something not obvious in raw numbers.
- **Question:** "What pattern or outlier do you see here that you likely would have missed if you were looking at a standard data table?"

4. Actionable Use Cases
Goal: Determine if the visualization leads to new research questions.
- **Question:** "Based on the patterns shown in this visualization, what new hypothesis would you propose for further study?"

Synthesis Questionnaire
To wrap up the evaluation, you could ask:
- "In what ways did the color choices or labeling help or hinder your interpretation of the findings?"